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| **Duration** | 3 mins 9 secs |
| **Grade** | **15.00** out of 15.00 (**100**%) |

Question **1**

Complete

Mark 1.00 out of 1.00

In the context of sorting, what does the divide-and-conquer approach involve?

a. Rearranging data without sorting
b. Sorting data in a single pass
c. Sorting data sequentially
d. Dividing the input into parts, solving each part, and combining the solutions

The correct answer is: Dividing the input into parts, solving each part, and combining the solutions

Question **2**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

a. They are discarded
b. They are split again into smaller sublists
c. They are combined to form a single sorted list
d. They are compared element by element

The correct answer is: They are combined to form a single sorted list

Question **3**

Complete

Mark 1.00 out of 1.00

What type of problems can sorting help solve efficiently?

a. All of the above
b. Duplicates
c. Selection
d. Searching

The correct answer is: All of the above

Question **4**

Complete

Mark 1.00 out of 1.00

What does the Bubble Sort algorithm primarily focus on during each pass?

- a. Bubbling up the largest element to its correct position
- b. Bubbling up the smallest element
- c. Dividing the list into halves
- d. Sorting the entire list in one pass

The correct answer is: Bubbling up the largest element to its correct position

Question **5**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is based on repeatedly dividing the list into halves?

- a. Insertion Sort
- b. Quick Sort
- c. Merge Sort
- d. Bubble Sort

The correct answer is: Merge Sort

Question **6**

Complete

Mark 1.00 out of 1.00

What is a key disadvantage of Bubble Sort compared to more advanced algorithms like Merge Sort?

- a. Bubble Sort does not guarantee sorted order
- b. Bubble Sort is difficult to implement
- c. Bubble Sort cannot handle duplicate elements
- d. Bubble Sort is less efficient for large lists

The correct answer is: Bubble Sort is less efficient for large lists

Question **7**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

a. Sorting

b. Merging

c. Complexity

d. Searching

The correct answer is:
Complexity

Question **8**

Complete

Mark 1.00 out of 1.00

Algorithm design technique used in merge sort algorithm is

a. Greedy method

b. Dynamic programming

c. Backtracking

d. Divide and conquer

The correct answer is:
Divide and conquer

Question **9**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

a. It only works on small datasets

b. It simplifies the sorting process by using only one pass

c. It allows for efficient parallel processing and sorting of data

d. It avoids the need for recursion

The correct answer is: It allows for efficient parallel processing and sorting of data

Question **10**

Complete

Mark 1.00 out of 1.00

How does Merge Sort achieve its efficiency?

a. By sorting data in a single pass
b. By breaking the input into smaller parts and merging them
c. By using the bubble-up method
d. By comparing elements sequentially

The correct answer is: By breaking the input into smaller parts and merging them

Question **11**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

a. Merge Sort
b. Linear Search
c. Bubble Sort
d. Binary Search

The correct answer is: Bubble Sort

Question **12**

Complete

Mark 1.00 out of 1.00

Which of the following is not an in-place sorting algorithm?

a. Heap sort
b. Selection sort
c. Merge sort
d. Quick sort

The correct answer is:
Merge sort

Question **13**

Complete

Mark 1.00 out of 1.00

What is one of the key advantages of using the built-in sorted() function in Python?

a. It requires external libraries
b. It only works with integer arrays
c. It is less efficient than custom sorting algorithms
d. It sorts data out of the box efficiently

The correct answer is: It sorts data out of the box efficiently

Question **14**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after dividing the input into smaller parts?

a. Each part is sorted independently
b. Each part is searched for a specific element
c. The parts are merged without sorting
d. The parts are ignored

The correct answer is: Each part is sorted independently

Question **15**

Complete

Mark 1.00 out of 1.00

Which of the following is a key reason for the importance of sorting algorithms?

a. Sorting decreases the efficiency of selection operations
b. Sorting helps in finding duplicates quickly
c. Sorting makes it harder to search for items
d. Sorting is rarely used in programming

The correct answer is: Sorting helps in finding duplicates quickly